Sentinel Auditor — Complete Project Report

Project Name: Sentinel Auditor (internal crate name: `needle`) **Submission:** Smart India Hackathon (SIH) 2026 **Deadline:** August 25, 2026 **Repository:** `d:\AEGIS_AST`

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1. Problem Statement

Government ministries (Defence, Finance, NIC), state enterprises, and regulatory bodies manage highly classified codebases powering critical infrastructure — tax processing, citizen ID systems, banking ledgers, defence communications. These codebases:

- Span **millions of lines** across legacy languages (COBOL, Fortran) and modern stacks (Rust, Python, Java).
- Must comply with **strict IT security policies** — token expiration limits, PII handling restrictions, cryptographic algorithm mandates — issued as internal circulars, PDFs, or formal directives.
- Are audited manually by security teams who read code file-by-file, cross-referencing policy clauses by hand.

Existing AI tools fail here because: 1. **Cloud dependency:** Tools like GitHub Copilot, Snyk, and SonarQube send code to external servers. Classified code **cannot leave the local network**. 2. **No policy awareness:** Static analysis tools flag generic CWE patterns but cannot ingest an organization's *specific* policy document and check code against *those exact clauses*. 3. **No audit trail:** There is no cryptographic proof that an audit was conducted or that the results haven't been tampered with after the fact.

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2. Proposed Solution

Sentinel Auditor is a sovereign, local-first code intelligence engine and AI policy auditor that:

1. **Runs entirely offline** — no cloud, no API keys, no data exfiltration. A compile-time `--features sovereign` flag structurally removes all networking code.
2. **Ingests security policy documents** (PDF, Markdown, plain text) and structures them into machine-readable obligations.
3. **Indexes codebases** using AST-level semantic chunking (via Tree-sitter) and builds both a keyword index (BM25) and a vector index (HNSW) — all hand-rolled in Rust.

4. **Cross-references policies against code** using a local LLM (Ollama) to find violations, with the query engine providing evidence.
5. **Produces a tamper-evident audit trail** — a SHA-256 hash-chained, Ed25519-signed JSONL ledger that proves audits happened and results haven't been altered.

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3. Approach & Design Philosophy

Principle	What It Means
Sovereign-first	Zero network capability in sovereign mode. Not "disabled by default" — the code is *compiled out*.
No external infrastructure	No Qdrant, Pinecone, ElasticSearch, PostgreSQL (in sovereign mode). Everything is a single binary.
Hand-rolled core	BM25, HNSW, and embeddings are built from scratch in Rust. No crate dependency for the search engine.
Semantic, not syntactic	We parse ASTs, not regex patterns. A "function" chunk contains its entire body, signature, and docstring.
Graceful degradation	If Tree-sitter doesn't support a language (COBOL, Fortran), we fall back to text chunking. Nothing is skipped.
Cryptographic accountability	Every audit report is hash-chained and digitally signed. Tampering is detected and localized to the exact block.

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4. Complete Tech Stack

Layer	Technology	Why
Language	Rust	Memory safety, zero-cost abstractions, no GC pauses, single static binary
AST Parsing	`tree-sitter` 0.20	Incremental parsing, multi-language support, mature ecosystem

Layer	Technology	Why
Language Grammars	`tree-sitter-python`, `-rust`, `-cpp`, `-typescript`, `-go`, `-java`, `-php`	Covers 7 languages natively
Content Hashing	`xxhash-rust` (xxH3, xxH64)	~30 GB/s throughput, used for deduplication and embeddings
Cryptographic Hashing	`sha2` (SHA-256)	Ledger block hashing
Digital Signatures	`ed25519-dalek`	Ledger block signing, non-repudiation
Serialization	`serde`, `serde_json`, `bincode`	JSON for human-readable data, Bincode for compact index persistence
PDF Parsing	`pdf-extract`	Policy document ingestion
CLI Framework	`clap` (derive)	Subcommand parsing with rich help text
Async Runtime	`tokio` (full)	Server, file watcher, LLM calls
Parallelism	`rayon`, `crossbeam`	Parallel chunking across files, channel-based watcher pipeline
File Watching	`notify`	OS-native inotify/ReadDirectoryChanges file watcher
Memory Mapping	`memmap2`	Memory-mapped file access for large codebases
HTTP Server	`axum` (cloud mode only)	REST API for the web UI
HTTP Client	`reqwest` (cloud mode only)	LLM API calls in cloud mode
Database	`rusqlite` (bundled) / `sqlx` (cloud)	Sessions, per-user API keys
Desktop Wrapper	`tauri` v2	Native window without Electron bloat

Layer	Technology	Why
Frontend	HTML/JS/CSS + React (Tauri UI)	Single-file SPA embedded at compile time
Graph Visualization	D3.js v7	Force-directed call graph, Sankey diagrams
Local LLM	Ollama (`qwen2.5-coder:7b-q4_0`)	Air-gapped inference on localhost
Unicode	`unicode-normalization`, `unicode-segmentation`	NFKD normalization for tokenization
Progress Bars	`indicatif`	Terminal progress during indexing
Colored Output	`colored`	Rich CLI output
Error Handling	`thiserror`, `anyhow`	Typed errors with context propagation
Logging	`tracing`, `tracing-subscriber`	Structured logging with env-filter

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5. Repository Layout

```
d:\AEGIS_AST\
├── Cargo.toml                    # Workspace root, dependencies, feature flags
├── src/
│   ├── main.rs                   # CLI entry point (clap subcommands)
│   ├── lib.rs                    # Library crate root, module exports
│   ├── error.rs                  # Error enum (14 variants)
│   ├── schema.rs                 # Core types: Chunk, Language, ChunkType, SearchR
│   ├── config.rs                 # Configuration loading
│   ├── chunking/
│   │   ├── mod.rs                # Chunker trait + language→chunker dispatch
│   │   ├── code.rs               # Tree-sitter AST chunker (7 languages + fallback
│   │   └── prose.rs              # Markdown heading / plain-text paragraph chunker
│   ├── indexing/
│   │   ├── bm25.rs               # Hand-rolled BM25 inverted index
```

```
│   │   └── hnsw.rs               # Hand-rolled HNSW approximate nearest neighbor g
│   ├── embedding/
│   │   └── mod.rs                # Hash-projection 384-dim embeddings (+ Ollama fa
│   ├── query/
│   │   ├── mod.rs                # QueryEngine orchestrator
│   │   └── fusion.rs             # Reciprocal Rank Fusion merger
│   ├── graph/
│   │   └── mod.rs                # CodeGraph: nodes, edges, communities, god nodes
│   ├── storage/
│   │   └── mod.rs                # JSON/Bincode persistence to .needle/index/
│   ├── watcher/
│   │   └── mod.rs                # notify-based file watcher for live re-indexing
│   ├── llm.rs                    # LLM routing: Anthropic → OpenAI → Groq → Ollama
│   ├── server/                   # (cloud mode only)
│   │   ├── indexer.rs            # Background async worker for cloud repos
│   │   ├── index_pipeline.rs     # Core indexing pipeline (walkdir → chunk → embed
│   │   ├── users.rs              # PostgreSQL user management
│   │   └── oauth.rs              # GitHub OAuth flow
│   ├── policy/
│   │   ├── mod.rs                # Policy subsystem root
│   │   ├── parser.rs             # PDF/Markdown/text ingestion + scanned-PDF guard
│   │   ├── clause.rs             # Data models: PolicyDocument, PolicyClause, Poli
│   │   ├── structurer.rs         # LLM + rule-based obligation structuring
│   │   └── linker.rs             # Compliance matching, evidence, violation detect
│   ├── ledger/
│   │   ├── mod.rs                # Ledger subsystem root + append API
│   │   ├── block.rs              # LedgerBlock, EntryType, canonical JSON serializ
│   │   ├── crypto.rs             # SHA-256 hashing + Ed25519 signing primitives
│   │   ├── keypair.rs            # Keypair generation, storage, redacted Debug
│   │   └── verifier.rs           # Block validation + exact tamper localization
│   └── cli/
│       ├── mod.rs                # CLI command dispatch
│       ├── serve/mod.rs          # HTTP server startup + all API route registratio
│       ├── audit.rs              # `sentinel audit` command
│       ├── ledger.rs             # `sentinel ledger` command
│       ├── doctor.rs             # `sentinel doctor` command
│       ├── policy.rs             # `sentinel policy` command
│       └── mcp/mod.rs            # MCP server (21 tools over stdio/JSON-RPC)
├── src-tauri/                    # Tauri v2 desktop app wrapper
│   └── tauri.conf.json           # App identity, window config, bundled resources
├── ui/                           # Frontend (Vite + React + TypeScript)
│   ├── index.html                # Main SPA entry
```

```
│   └── src/App.tsx                # React app with 4 tabs
├── needle-vscode/                 # VS Code extension
│   └── package.json               # Commands, views, configuration
├── demo/                          # Demo assets for live presentation
│   ├── finance_ministry_data_policy.md
│   ├── vulnerable_auth.py
│   └── legacy_system.cobol
├── scripts/                       # Automation scripts
│   ├── run_policy_demo.ps1
│   ├── warmup_demo.ps1
│   ├── eval_precision_recall.py
│   └── benchmark_latency.ps1
├── docs/
│   ├── 01_PRD.md
│   ├── 02_TECH_STACK_AND_FLOWS.md
│   ├── 03_SCHEMA.md
│   ├── 04_DESIGN.md
│   └── BENCHMARKS.md
├── ARCHITECTURE.md
├── PROJECT.md
└── TEST_INFRA.md
```

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6. Data Model — The `Chunk` and Its Friends

The `Chunk` is the **atomic unit** of everything in Sentinel Auditor. Every file is broken into chunks, every chunk is indexed, embedded, searched, and returned.

`Chunk` Struct (defined in `schema.rs`)

Field	Type	Purpose
`id`	`u64`	Unique monotonic identifier
`file_path`	`String`	Absolute path to the source file
`root_dir`	`u16`	Index of the watched root directory
`byte_offset`	`u64`	Starting byte position in the file

Field	Type	Purpose
`byte_length`	`u32`	Length of the chunk in bytes
`line_start`	`u32`	Starting line number (1-indexed)
`line_end`	`u32`	Ending line number (inclusive)
`language`	`Language`	Detected programming language
`chunk_type`	`ChunkType`	Semantic category (Function, Class, etc.)
`content_hash`	`u64`	xxH3-64 hash of the trimmed content (for dedup)
`token_count`	`u32`	Whitespace-split token count
`embedding_id`	`u64`	Pointer into the HNSW embedding store
`status`	`ChunkStatus`	`Active` or `Tombstoned`
`created_at`	`u64`	Unix timestamp of creation
`tombstoned_at`	`Option<u64>`	Timestamp of soft-deletion
`content`	`String`	The actual text content of the chunk

`Language` Enum — 17 Variants

`Rust`, `Python`, `TypeScript`, `JavaScript`, `Go`, `Java`, `C`, `Cpp`, `Markdown`, `PlainText`, `Toml`, `Yaml`, `Json`, `Dockerfile`, `Shell`, `Pdf`, `Php`.

`ChunkType` Enum — 9 Variants

`Function`, `Method`, `Class`, `Module`, `Import`, `Comment`, `Paragraph`, `Section`, `ConfigBlock`.

Other Key Types

Type	Fields	Purpose
`PostingsEntry`	`chunk_id: u64`, `term_freq: u16`	BM25 inverted index entry

Type	Fields	Purpose
`SearchResult`	`chunk_id`, `file_path`, `line_start`, `line_end`, `language`, `chunk_type`, `content`, `score: f32`, `signals: SearchSignal`	Returned to the UI
`SearchSignal`	`Keyword`, `Semantic`, `Hybrid`	Which index produced this result
`WalEntry`	`ChunkAdded(Chunk)`, `ChunkDeleted(u64)`, `FileModified{...}`, `Checkpoint(u64)`	Write-ahead log entries
`IndexMetadata`	`version`, `total_chunks`, `total_files`, `avg_chunk_length`, `hnsw_m`, `hnsw_ef_construction`, `bm25_k1`, `bm25_b`, `embedding_dim`, etc.	Persisted index stats

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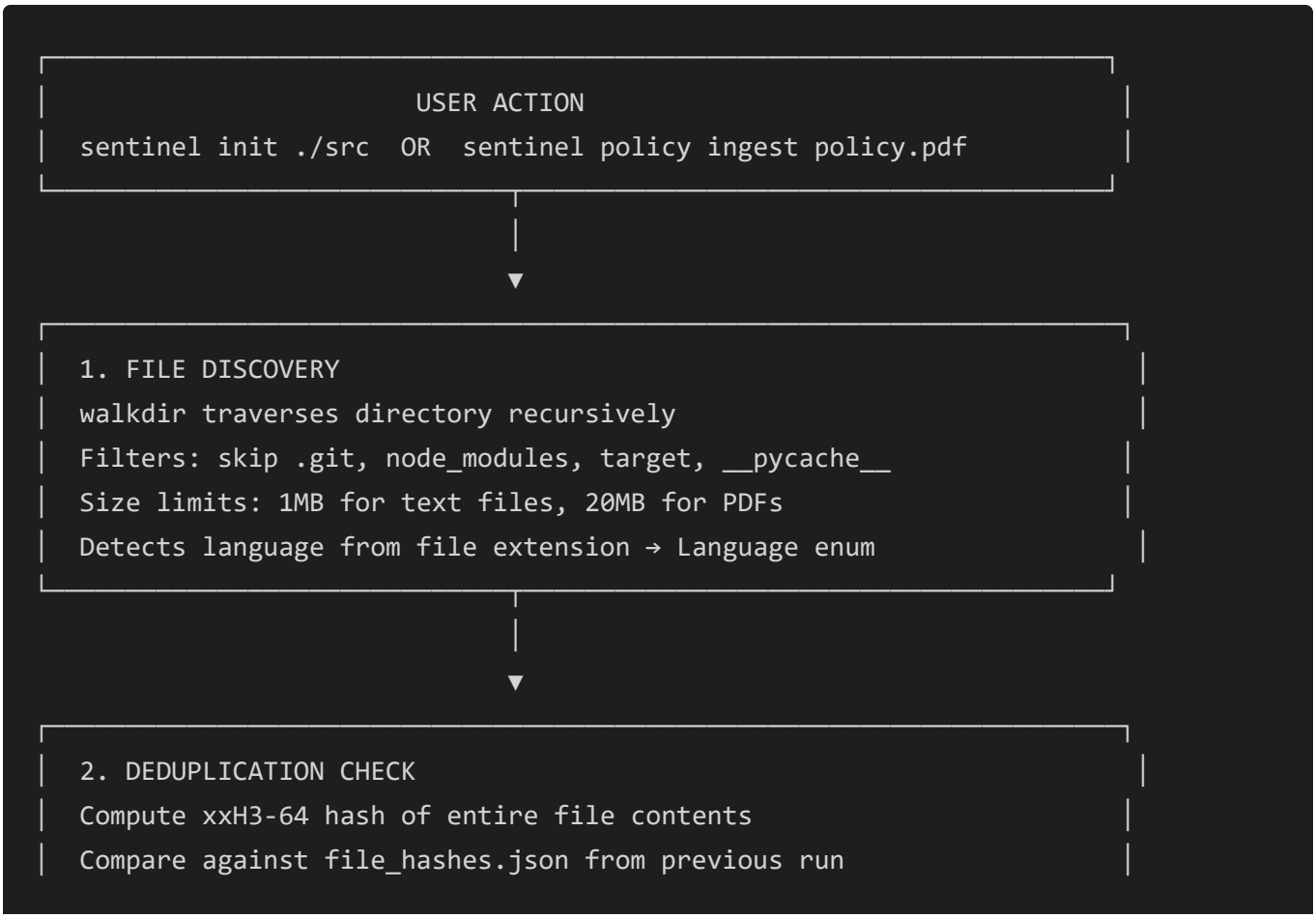
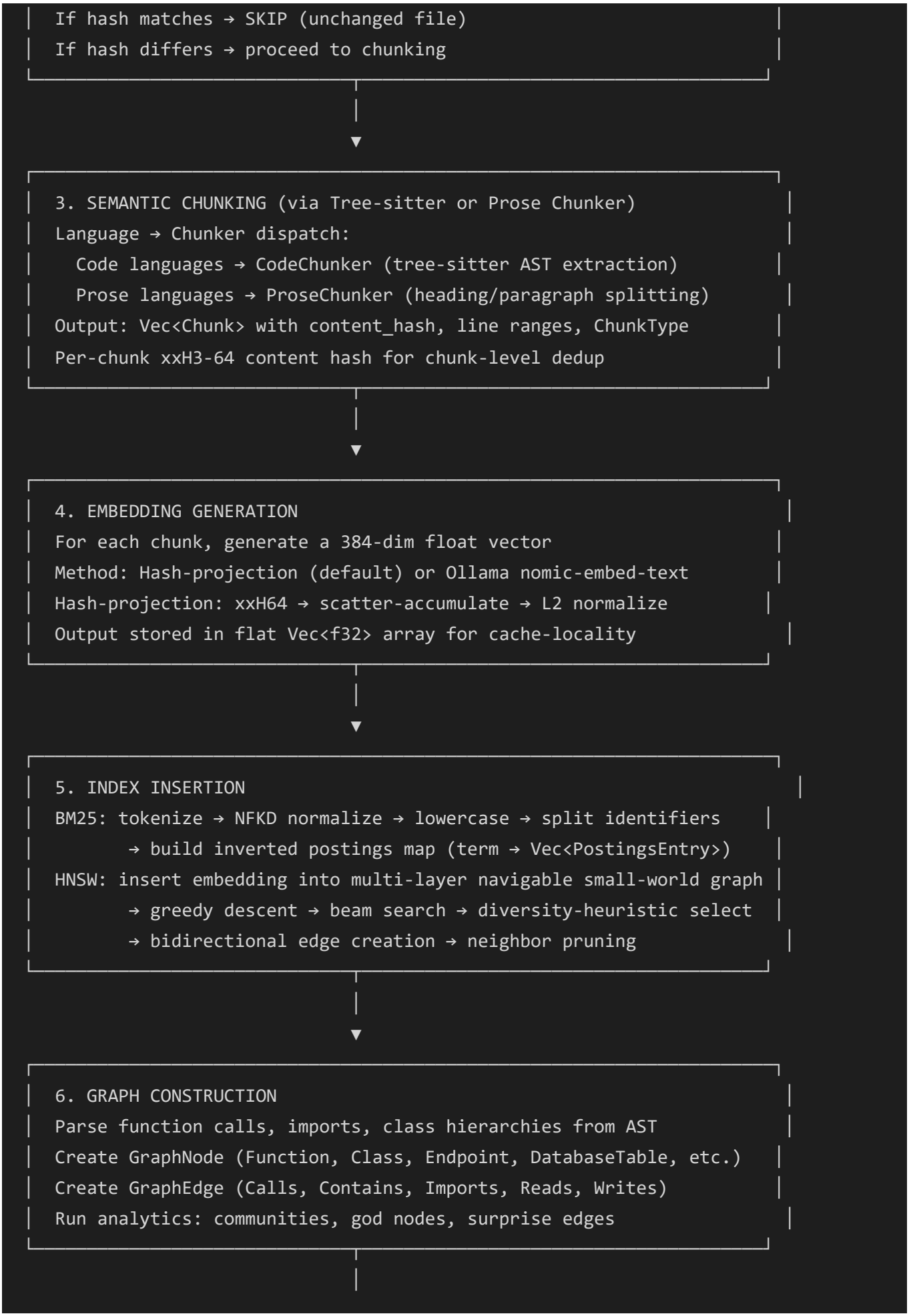
7. Data Flow: End-to-End Pipeline




If hash matches → SKIP (unchanged file)
If hash differs → proceed to chunking
3. SEMANTIC CHUNKING (via Tree-sitter or Prose Chunker)
Language → Chunker dispatch:
Code languages → CodeChunker (tree-sitter AST extraction)
Prose languages → ProseChunker (heading/paragraph splitting)
Output: Vec<Chunk> with content_hash, line ranges, ChunkType
Per-chunk xxH3-64 content hash for chunk-level dedup
4. EMBEDDING GENERATION
For each chunk, generate a 384-dim float vector
Method: Hash-projection (default) or Ollama nomic-embed-text
Hash-projection: xxH64 → scatter-accumulate → L2 normalize
Output stored in flat Vec<f32> array for cache-locality
5. INDEX INSERTION
BM25: tokenize → NFKD normalize → lowercase → split identifiers
→ build inverted postings map (term → Vec<PostingsEntry>)
HNSW: insert embedding into multi-layer navigable small-world graph
→ greedy descent → beam search → diversity-heuristic select
→ bidirectional edge creation → neighbor pruning
6. GRAPH CONSTRUCTION
Parse function calls, imports, class hierarchies from AST
Create GraphNode (Function, Class, Endpoint, DatabaseTable, etc.)
Create GraphEdge (Calls, Contains, Imports, Reads, Writes)
Run analytics: communities, god nodes, surprise edges

```
                          ▼
┌──────────────────────────────────────────────────────────────────┐
│  7. PERSISTENCE                                                  │
│  .needle/index/chunks.json    — all chunks as JSON               │
│  .needle/index/bm25.bin       — BM25 index as Bincode            │
│  .needle/index/hnsw.bin       — HNSW graph + embeddings as Bincode   │
│  .needle/index/meta.json      — index metadata                   │
│  .needle/index/graph.json     — full code graph                  │
│  .needle/index/filemap.json   — path → chunk_id mapping          │
│  .needle/index/file_hashes.json — path → xxH3 hash (for dedup)   │
│  .needle/policy/              — ingested PolicyDocument JSON files   │
└──────────────────────────────────────────────────────────────────┘
```

Query-Time Data Flow

```
User types query → Embed query (hash-projection, 0.00ms)
                  ├→ BM25 search (tokenize → IDF-weighted scoring)
                  └→ HNSW search (beam search at layer 0)
                       │
                       ▼
              Reciprocal Rank Fusion (k=60)
              Merge and re-rank results
                       │
                       ▼
              Deduplication (50% line overlap)
              Language filtering
                       │
                       ▼
              Return Vec<SearchResult> to UI/MCP/CLI
```

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8. Deep Dive: Tree-sitter AST Chunking

What Is Tree-sitter?

Tree-sitter is an incremental parsing library that builds a concrete syntax tree from source code. Unlike regex-based extractors, it understands the actual grammar of the language.

How We Use It

The `CodeChunker` (in `src/chunking/code.rs`) parses each file into an AST and walks it with a `TreeCursor` (non-recursive DFS to avoid stack overflows on deeply nested code).

AST Node Types Extracted Per Language

Language	Node Types → ChunkType
Rust	`function_item` → Function, `struct_item` → Class, `enum_item` → Class, `trait_item` → Class, `impl_item` → Class, `macro_definition` → Function
Python	`function_definition` → Function, `class_definition` → Class
TypeScript/JS	`function_declaration` → Function, `function_expression` → Function, `arrow_function` → Function, `method_definition` → Method, `class_declaration` → Class, `export_statement` → Module
Go	`function_declaration` → Function, `method_declaration` → Method, `type_declaration` → Class
Java	`method_declaration` → Method, `class_declaration` → Class
PHP	`function_definition` → Function, `method_declaration` → Method, `class_declaration` → Class
C/C++	`function_definition` → Function, `struct_specifier` → Class, `class_specifier` → Class

Extraction Algorithm (Step by Step)

1. Initialize `tree_sitter::Parser` with the appropriate language grammar.
2. Parse the source code into a `Tree`.
3. Walk the tree using a `TreeCursor` (DFS, non-recursive):
4. At each node, check if its `kind()` matches the target node types for the language.
5. If matched: extract `start_byte`, `end_byte`, `start_position().row` (for line numbers), and the text slice.
6. If the node is a Class/Impl: descend into children to also extract internal Methods.
7. Otherwise: skip the subtree entirely (don't descend into function bodies looking for nested functions).
8. **"God function" splitting**: If a function exceeds **150 lines**, subdivide it:
9. Try to split at blank lines or block comments.

10. If that fails, force-split every **100 lines**.
11. Filter out chunks with fewer than 3 whitespace-separated tokens.
12. For each surviving chunk, compute `content_hash = xxh3_64(text.as_bytes())`.

Fallback Strategy

If Tree-sitter fails (returns `None` from `extract_blocks`) OR the language has no grammar: - **Fallback line-window chunker** activates. - Window size: **60 lines**, step size: **45 lines** (creating 15 lines of overlap between consecutive chunks). - All fallback chunks get `ChunkType::Module`.

Prose Chunking (Markdown / Plain Text)

For non-code files: - **Markdown**: Split on ATX headings (`#`). Each heading + its body becomes a `Section` chunk. - **Plain text**: Split on double-newlines (`\n\n`). Each paragraph becomes a `Paragraph` chunk. Paragraphs with fewer than 3 words are skipped.

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9. Deep Dive: BM25 Inverted Index (Hand-Rolled)

What Is BM25?

BM25 (Okapi Best Match 25) is a probabilistic ranking function for keyword search. It scores documents based on term frequency, inverse document frequency, and document length normalization.

Our Implementation (`src/indexing/bm25.rs`)

Data Structures

```
BM25Index {
    postings:       HashMap<String, Vec<PostingsEntry>>  // term → [(chunk_id, tf)]
    doc_freqs:      HashMap<String, u32>                  // term → number of docs c
    chunk_lengths:  HashMap<u64, u32>                      // chunk_id → token count
    deleted_chunks: HashSet<u64>                           // soft-deleted chunk IDs
    total_docs:     u64                                    // total document count
    avg_doc_length: f32                                    // running average (Welfo
}
```

Parameters

- **k1 = 1.2** — controls term frequency saturation. Higher values give more weight to repeated terms.
- **b = 0.75** — controls document length normalization. 1.0 = full normalization, 0.0 = no normalization.

Tokenization Pipeline

1. Apply **NFKD Unicode normalization** (decomposes accented characters).
2. **Lowercase** the entire string.
3. Split on **non-alphanumeric characters** (preserving underscores `_`).
4. Filter out **stopwords** (common English words: "the", "is", "at", "in", etc.).
5. **Identifier splitting**: For `snake_case` identifiers like `retry_with_backoff`, emit both the whole identifier AND the individual parts (`retry`, `with`, `backoff`). This allows matching on partial identifier names.

Indexing Algorithm (`add_chunk`)

1. Tokenize the chunk's content.
2. Count per-term frequencies (TF) using a local `HashMap`.
3. For each unique term, append a `PostingsEntry { chunk_id, term_freq }` to the inverted `postings` map.
4. Increment `doc_freqs[term]` by 1.
5. Update `avg_doc_length` using **Welford's online mean** algorithm (numerically stable incremental update).

Search Algorithm (`search`)

1. Tokenize the query.
2. For each query term:
3. Look up the postings list.
4. Calculate **IDF** using the Robertson-Sparck Jones formula: `idf = ln((N - df + 0.5) / (df + 0.5) + 1.0).max(0.0)` where `N = total_docs`, `df = doc_freq`.
5. For each posting entry (skipping deleted chunks):
    - Fetch document length `dl = chunk_lengths[chunk_id]`.
    - Compute TF normalization: `tf_norm = (tf × (k1 + 1)) / (tf + k1 × (1 - b + b × dl / avg_dl))`
    - Add `idf × tf_norm` to the chunk's cumulative score.

6. Sort all scored chunks by score descending.
7. Return top `limit` results.

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10. Deep Dive: HNSW Vector Index (Hand-Rolled)

What Is HNSW?

Hierarchical Navigable Small World is a state-of-the-art algorithm for Approximate Nearest Neighbor (ANN) search. It builds a multi-layered graph where each layer is a navigable small-world network, with higher layers containing fewer, more spread-out nodes for fast coarse navigation, and lower layers containing all nodes for fine-grained search.

Our Implementation (`src/indexing/hnsw.rs`)

Configuration Constants

Parameter	Default	Purpose
`M`	16	Max connections per node per layer (except layer 0)
`M_max0`	32 (= 2 × M)	Max connections at layer 0 (the densest layer)
`ef_construction`	200	Beam width during insertion (higher = more accurate, slower build)
`ml`	`1/ln(M)` ≈ 0.36	Level generation multiplier
Max layers	16	Hard cap on graph depth

Data Structure

```
HnswIndex {
    embeddings:  Vec<f32>           // Flat array, dim=384, contiguous for cache loc
    nodes:       HashMap<u64, HnswNodeData>   // chunk_id → per-layer adjacency list
    dim:         usize              // 384
    entry_point: Option<u64>        // Global entry node (highest-layer node)
    max_layer:   usize              // Current maximum layer in the graph
```

```
    deleted:     HashSet<u64>       // Soft-deleted nodes (tombstones)
}

HnswNodeData {
    layer:      usize                          // Node's assigned layer
    neighbors:  Vec<Vec<u64>>                  // neighbors[layer_i] = [node_ids...
}
```

Distance Metric

Cosine distance: `distance = 1.0 - dot(a, b)` where vectors are pre-L2-normalized during embedding generation, so dot product equals cosine similarity.

Insertion Algorithm (`add_node`)

1. **Layer assignment**: Sample from geometric distribution: `layer = floor(-ln(random()) × ml)`, capped at 16.
2. **Phase 1 — Greedy descent** (from `max_layer` down to `node_layer + 1`):
3. Start at the global `entry_point`.
4. At each layer, greedily move to the nearest neighbor (ef=1).
5. This quickly narrows down to the right neighborhood.
6. **Phase 2 — Beam search + connect** (from `node_layer` down to 0):
7. Run beam search with `ef_construction = 200` candidates.
8. Select `M` neighbors using the **diversity heuristic**: reject a candidate if it's closer to an already-selected neighbor than to the query. This prevents clustering and ensures diverse connections.
9. Create **bidirectional edges** between the new node and selected neighbors.
10. **Prune**: If any neighbor exceeds `M_max0` (layer 0) or `M` (higher layers) connections, remove the longest edge.
11. If the new node's layer exceeds `max_layer`, update `entry_point` and `max_layer`.

Search Algorithm (`search_knn`)

1. **Phase 1 — Greedy descent** (from `max_layer` to layer 1): ef=1, find the entry point for layer 0.
2. **Phase 2 — Beam search** at layer 0:
3. Maintain a min-heap of candidates and a max-heap of results.
4. Expand the closest unvisited candidate's neighbors.

5. Skip tombstoned nodes.
6. Continue until no closer unvisited candidates remain.
7. Return top `k` results sorted by distance.

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11. Deep Dive: Hash-Projection Embeddings

Why Not Use a Neural Model?

Traditional embedding models (MiniLM, ONNX transformers) require: - A model file download (50-100MB). - An inference runtime (ONNX Runtime or libtorch). - Significant memory and latency per chunk.

We use a **hash-projection** approach that generates 384-dim vectors in **0.00ms** with zero external dependencies.

Algorithm (`src/embedding/mod.rs`)

1. Split the input text into whitespace tokens.
2. For each token, get its bytes.
3. Compute three hashes using `xxh64` with different seeds:
4. `h0 = xxh64(bytes, seed=0)` → position 1
5. `h1 = xxh64(bytes, seed=1)` → position 2
6. `h2 = xxh64(bytes, seed=2)` → sign determination
7. Accumulate into a 384-dimensional vector: `sign0 = if h2 & 1 == 0 { 1.0 } else { -1.0 } sign1 = if (h2 >> 1) & 1 == 0 { 1.0 } else { -1.0 } acc[h0 % 384] += sign0 acc[h1 % 384] += sign1`
8. **L2-normalize** the final vector so `||v|| = 1.0`.
9. This enables dot-product to serve as cosine similarity (since both vectors are unit-normalized).

Why This Works

This is a form of **random projection** (Johnson-Lindenstrauss lemma): high-dimensional data can be projected into lower dimensions while approximately preserving pairwise distances. The xxH64 hash acts as a pseudo-random hash function, and the sign bits provide directional diversity.

Alternative: Ollama Embeddings

When running with Ollama, the system can optionally use `nomic-embed-text` for higher-quality neural embeddings via `POST http://localhost:11434/api/embeddings`. The hash-projection is the default for sovereign/offline deployments.

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12. Deep Dive: Reciprocal Rank Fusion (RRF)

What Is RRF?

RRF is a rank aggregation algorithm that merges results from multiple independent rankers without needing their raw scores (which may be on different scales).

Our Implementation (`src/query/fusion.rs`)

Formula

```
rrf_score(d) = Σ  1 / (k + rank_i(d))
               i
```

where `k = 60` (a constant that prevents top-ranked documents from dominating).

Algorithm (Step by Step)

1. Receive ranked lists from BM25 and HNSW.
2. For each result in the BM25 list at rank `r`: `scores[chunk_id] += 1.0 / (60.0 + r)`
3. For each result in the HNSW list at rank `r`: `scores[chunk_id] += 1.0 / (60.0 + r)`
4. Sort all chunk IDs by their cumulative RRF score, descending.
5. Apply **50% line-overlap deduplication**: if two chunks from the same file overlap by more than 50% of their line ranges, keep only the higher-scored one.
6. Apply optional language filter.
7. Return the final merged list.

Why RRF?

- BM25 returns scores on an unbounded log scale (0 to ~30).
- HNSW returns cosine distances (0 to 2).
- RRF doesn't need to normalize these — it only uses rank position, making it robust to score-scale differences.

13. Deep Dive: CodeGraph — Call Graph, Communities, and God Nodes

What It Does

The `CodeGraph` (in `src/graph/mod.rs`) builds a rich knowledge graph of the codebase's structure: which functions call which, which modules import what, which endpoints exist, and which database tables are accessed.

Node Types

Kind	Source	Example
`Module`	File-level scope	`src/query/mod.rs`
`Function`	AST function definitions	`fn search_knn()`
`Method`	AST method definitions	`fn add_chunk(&mut self)`
`Class` / `Struct` / `Trait`	AST type definitions	`struct BM25Index`
`Endpoint`	HTTP route detection	`GET /api/search`
`DatabaseTable`	SQL string extraction	`users`, `sessions`
`GlobalState`	`static mut`, `window.*`	`CONFIG_INSTANCE`

Edge Types

Edge	Meaning	Detection
`Contains`	Module → Definition	File scope containment
`Imports`	File → File	`use`, `import`, `require` statements
`Calls`	Function → Function	Function name in call expression AST nodes
`Reads`	Function → Database/Global	`SELECT FROM table` or global variable read

Edge	Meaning	Detection
`Writes`	Function → Database/Global	`INSERT INTO`, `UPDATE`, or global mutation

Endpoint Detection

- **Express.js**: Regex for `.get('/path', handler)`, `.post(...)`, `.put(...)`, `.delete(...)`.
- **Axum**: Regex for `.route("/path", get(handler))`.
- HTTP method extracted and stored as `detail` on the node.

Analytics Algorithms

Community Detection (Label Propagation)

1. Assign each node a unique label (its own ID).
2. Iterate up to **50 times**:
3. For each node, look at all neighbors connected via `Calls` or `Imports` edges.
4. Count the frequency of each neighbor's label.
5. Assign the most frequent label to this node.
6. Converge when no labels change.
7. Nodes sharing the same label form a **community** — a cluster of tightly related code.

God Node Detection

- Compute `degree = in_edges + out_edges` for every node.
- Sort by degree descending.
- The top nodes are "God Objects" — functions or classes that touch too many things, indicating a refactoring target.

Surprise Edge Detection

- Find edges that connect nodes in **different communities**.
- These "cross-community bridges" are architecturally interesting — they represent unexpected dependencies between otherwise isolated clusters.

---

14. Deep Dive: Policy Compliance Subsystem

Overview

This subsystem ingests security policy documents, structures them into machine-readable obligations, and cross-references those obligations against the indexed codebase to find violations.

Phase 1: Policy Ingestion (`src/policy/parser.rs`)

Supported Formats

Format	Parser
`.pdf`	`pdf_extract` crate
`.md`	Native Markdown reader
`.txt`	Raw text reader
`.policy`	Custom DSL format

Scanned-PDF Guard

After extracting text from a PDF, the parser counts **printable, non-whitespace, non-control characters**. If `printable_chars < 20`, it means the PDF is likely a scanned image with no extractable text. The system returns a **loud error**: `PolicyError("Scanned or image-only PDF detected. Use OCR preprocessing...")` rather than silently indexing an empty document.

Clause Chunking Algorithm

The parser splits the policy document into individual clauses: 1. Iterate through lines. 2. A new clause starts when a line matches: - Markdown ATX headings (`#`, `##`, `###`) - Section symbols (`§`) - Keywords: `Section`, `Article`, `Clause`, `Requirement`, `Rule`, `Policy` - Decimal numbering: `1.1`, `2.3.1` - Lettered sections: `A.`, `B.` 3. If no headers are detected, fall back to splitting on double-newlines.

Each clause gets an auto-generated `clause_number`, `line_start`, `line_end`, `byte_offset`, and `byte_length`.

Phase 2: Obligation Structuring (`src/policy/structurer.rs`)

Each clause is decomposed into concrete **obligations** — machine-readable rules.

LLM-based Structuring (when Ollama is available)

A system prompt instructs the LLM to output JSON matching the `LlmObligationItem` schema:

```
{
  "title": "Token expiry limit",
  "description": "Authentication tokens must expire within 15 minutes",
  "obligation_type": "Must",
  "severity": "High",
  "target_entities": ["authentication", "token"],
  "action": "Enforce 15-minute token expiry",
  "condition": "For all citizen-facing applications",
  "target_keywords": ["JWT", "session", "token", "expiry"],
  "semantic_query": "token expiration time limit"
}
```

Rule-based Structuring (Heuristic Fallback)

If the LLM is unavailable: 1. Split the clause text into sentences. 2. For each sentence, detect: - **Modal verbs**: `must not`, `shall not` → `MustNot`; `must`, `shall` → `Must`; `should` → `Should`; `may` → `May`. - **Conditional triggers**: `if`, `when`, `where`, `unless` → `RequiredIf` / `ProhibitedIf`. 3. **Severity calculation**: Scan for critical keywords: - `password`, `private key`, `sql injection`, `remote code execution` → `Critical` / `High` - `deprecated`, `legacy` → `Medium` - Default based on obligation type: `MustNot` → `High`, `Must` → `Medium`, `Should` → `Low`. 4. **Target entity inference**: Scan for keywords like `route`, `handler`, `api` → `endpoint`; `function`, `method` → `function`; `struct`, `class` → `struct`.

Phase 3: Compliance Matching (`src/policy/linker.rs`)

Matching Algorithm

For each obligation: 1. Build a composite query from: `title + description + action + condition + semantic_query + target_keywords`. 2. Execute this query against the `QueryEngine` (BM25 + HNSW + RRF). 3. Keep results with `score > 0.01` (top 5).

Violation Detection

For each matched code chunk: - **Numeric analysis**: Extract numbers with units from both the obligation text and the code (e.g., `"15 minutes"` vs `"86400"` seconds). If the code value is `< 95%` of the required value for the same unit category, mark as `Violated`. - **Pattern matching**: If the obligation is `MustNot` and the code contains the prohibited pattern, mark as `Violated`.

Output Data Structures

Type	Fields
`ComplianceStatus`	`Satisfied { evidence }`, `Violated { conflicting, reason }`, `NoImplementationFound`
`EvidenceNode`	`chunk_id`, `file_path`, `line_start`, `line_end`, `confidence` (RRF score), `snippet`
`ComplianceLink`	`obligation_id`, `clause_id`, `clause_number`, `title`, `obligation_type`, `severity`, `status`

---

15. Deep Dive: Cryptographic Audit Ledger

Purpose

To provide **tamper-evident, non-repudiable proof** that an audit was conducted and the results haven't been altered. Required for compliance reporting to external regulators.

Architecture (`src/ledger/`)

`LedgerBlock` Struct

Field	Type	Purpose
`sequence`	`u64`	0-based strictly monotonic counter
`timestamp`	`String`	RFC 3339 UTC timestamp
`prev_hash`	`String`	SHA-256 of the previous block (or 64 zeroes for genesis)

Field	Type	Purpose
`entry_type`	`EntryType`	Category of audit event
`payload_hash`	`String`	SHA-256 of the canonical JSON payload
`payload`	`serde_json::Value`	The actual audit report data
`signer_public_key`	`String`	64-char hex Ed25519 public key
`signature`	`String`	128-char hex Ed25519 signature
`block_hash`	`String`	SHA-256 of the complete block preimage

`EntryType` Enum

`ComplianceAudit`, `SecurityScan`, `PolicyIngest`, `CodebaseSnapshot`, `SystemEvent`.

Canonical JSON Serialization

To ensure deterministic hashing regardless of key insertion order: 1. Recursively traverse the JSON value. 2. For objects: store keys in a `BTreeMap` (guaranteed lexicographic order). 3. Serialize the `BTreeMap` to bytes. 4. This guarantees that the same payload always produces the same hash.

Block Creation Pipeline

1. **Hash the payload**: Canonicalize the JSON payload → SHA-256 → `payload_hash`.
2. **Construct signing preimage**: `"{sequence}:{timestamp}:{prev_hash}:{entry_type:?}:{payload_hash}"`.
3. **Sign**: Ed25519 sign the preimage bytes → `signature` (128-char hex).
4. **Construct block preimage**: `"{signing_preimage}:{signer_public_key}:{signature}"`.
5. **Hash the block**: SHA-256 of the block preimage → `block_hash`.
6. **Append**: Serialize the `LedgerBlock` as a single JSON line and append to `.needle/ledger/audit_chain.jsonl`.

Keypair Management

- **Generation**: Uses OS-provided CSPRNG (`OsRng`) to generate an Ed25519 `SigningKey`.

- **Storage**: Private key saved as hex to `.needle/ledger/key.priv` (Unix permissions `0o600`). Public key to `.needle/ledger/key.pub`.
- **Security**: `Debug` and `Display` traits on `LedgerKeypair` are overridden to print `"[REDACTED PRIVATE KEY]"` — the private key never appears in logs, even at debug level.

Verification Algorithm (Step by Step)

For each line in `audit_chain.jsonl`: 1. Parse JSON into `LedgerBlock`. 2. **Sequence check**: `block.sequence == expected_sequence` (must be strictly monotonic starting from 0). 3. **Chain link check**: `block.prev_hash == running_prev_hash` (must match the hash of the previous block). 4. **Payload integrity**: Canonicalize `block.payload` → SHA-256 → compare with `block.payload_hash`. 5. **Signature verification**: Reconstruct the signing preimage → verify the Ed25519 `signature` using the `signer_public_key`. 6. **Block hash integrity**: Reconstruct the block preimage → SHA-256 → compare with `block.block_hash`. 7. Update `running_prev_hash = block.block_hash` for the next iteration.

Tamper Localization

If **any** step fails, verification immediately halts and returns:

```
"TAMPER DETECTED at sequence {N}: {reason}"
```

where `{reason}` is one of: `sequence mismatch`, `prev_hash mismatch`, `payload_hash mismatch`, `signature verification failed`, `block_hash mismatch`.

---

16. Deep Dive: Local LLM Routing & Sovereign Mode

LLM Routing Priority (`src/llm.rs`)

1. `ANTHROPIC_API_KEY` → Anthropic Claude API
2. `OPENAI_API_KEY` → OpenAI API
3. `GROQ_API_KEY` → Groq API
4. **Ollama** → `http://localhost:11434` (always available locally)

In **sovereign mode**, steps 1-3 are compiled out. Only Ollama remains.

Zero-Dependency Ollama Client

In sovereign mode, even `reqwest` (the HTTP client crate) is compiled out. Instead, we implement a raw `LoopbackHttpClient` that: 1. Opens a `tokio::net::TcpStream` directly to `127.0.0.1:11434`. 2. Manually constructs HTTP/1.1 POST requests (headers, Content-Length, body). 3. Sends to `/api/chat` or `/api/generate`. 4. Manually parses the HTTP response: status line, headers, chunked transfer encoding, JSON body.

`--offline-strict` Flag

When enabled, the `LoopbackValidator` enforces: - Parse the target hostname. - Verify it is **exactly** `localhost`, `127.0.0.1`, `::1`, or in the `127.0.0.0/8` subnet. - **Non-loopback IPs or domain names** instantly trigger an `OfflineStrictViolation` error — **without performing a DNS query** (to prevent DNS-based exfiltration).

Sovereign Build Mode

The `Cargo.toml` defines:

```
[features]
default = ["cloud"]
cloud = ["axum", "tower-http", "open", "sqlx", "tower-cookies", "time", "urlencoding
sovereign = []
```

Building with `--no-default-features --features sovereign` completely removes all networking crate code from the binary. Verified via `cargo tree` showing zero networking dependencies.

---

17. Storage & Persistence Layer

All data lives under `.needle/` in the project root.

Path	Format	Contents
`.needle/index/chunks.json`	JSON	`HashMap<u64, Chunk>` — all chunks with their content
`.needle/index/bm25.bin`	Bincode	Serialized `BM25Index` (postings, doc_freqs, chunk_lengths)

Path	Format	Contents
`.needle/index/hnsw.bin`	Bincode	Serialized `HnswIndex` (flat embedding Vec + node adjacency)
`.needle/index/meta.json`	JSON	`IndexMetadata` (version, counts, HNSW/BM25 parameters)
`.needle/index/graph.json`	JSON	Serialized `CodeGraph` (nodes, edges, communities)
`.needle/index/filemap.json`	JSON	`path → Vec<chunk_id>` mapping
`.needle/index/file_hashes.json`	JSON	`path → u64` xxH3 hash (for incremental dedup)
`.needle/policy/*.json`	JSON	Individual `PolicyDocument` objects
`.needle/ledger/audit_chain.jsonl`	JSONL	Append-only cryptographic ledger (one block per line)
`.needle/ledger/key.priv`	Hex	Ed25519 private key (permissions 0o600)
`.needle/ledger/key.pub`	Hex	Ed25519 public key

Why Bincode for indices? JSON is human-readable but slow to parse for large structures. Bincode is a compact binary format that's ~10x faster to deserialize, critical for the BM25 postings map and HNSW adjacency lists which can contain millions of entries.

---

18. File Watcher — Live Re-indexing

The `FileWatcher` (`src/watcher/mod.rs`) uses the `notify` crate with `RecommendedWatcher` (which auto-selects the OS-native mechanism: `inotify` on Linux, `ReadDirectoryChangesW` on Windows, `FSEvents` on macOS).

Events Handled

OS Event	Internal Event	Action
`EventKind::Create`	`FileEvent::Created(path)`	Chunk + index the new file
`EventKind::Modify`	`FileEvent::Modified(path)`	Re-chunk, tombstone old chunks, index new ones
`EventKind::Remove`	`FileEvent::Deleted(path)`	Tombstone all chunks for this file

Events flow through a `crossbeam::channel` to the index pipeline running on a separate thread.

---

19. HTTP API — Complete Route Map

All routes are registered in `src/cli/serve/mod.rs` and served via Axum on port `7700`.

Method	Path	Purpose
`GET`	`/api/mode`	Server capability metadata (cloud vs sovereign)
`GET`	`/api/search?q=&limit=`	Hybrid BM25+HNSW search
`GET`	`/api/status`	Index stats and DB metadata
`POST`	`/api/open`	Open local file in the user's editor
`POST`	`/api/ask`	LLM RAG — ask questions with code context
`POST`	`/api/similar`	Find semantically similar chunks
`GET`	`/api/todos`	Scan for TODO/FIXME/HACK annotations
`GET`	`/api/files`	List files by language and chunk count
`GET`	`/api/graph`	Full CodeGraph JSON (nodes + edges)
`GET`	`/api/blast-radius?file=`	Callers/callees expansion from a file
`GET`	`/api/health`	Server health diagnostics
`GET`	`/api/security`	AST-level security pattern scan

Method	Path	Purpose
`GET`	`/api/patterns`	Code pattern analysis
`GET`	`/api/git/churn`	Git history analytics (most-changed files)
`POST`	`/api/import/github`	Import a GitHub repository (cloud mode)
`POST`	`/api/import/local`	Import a local directory
`GET`	`/api/import/status`	Background import progress polling
`POST`	`/api/import/clear`	Clear imported indexes
`GET`	`/api/audit`	Retrieve policy audit results
`GET`	`/api/ledger`	Get cryptographic ledger entries
`GET`	`/api/ledger/verify`	Verify ledger integrity
`POST`	`/api/ledger/sign`	Sign a custom audit block
`GET`	`/api/doctor`	Diagnostics health check
`GET`	`/auth/github`	Start OAuth flow (cloud mode)
`GET`	`/auth/callback`	OAuth callback handler
`GET`	`/auth/logout`	End session
`GET`	`/api/me`	Current user profile
`POST`	`/api/me/regenerate-key`	Regenerate API key
`POST`	`/api/me/revoke-key`	Revoke API key
`POST`	`/api/validate-key`	Validate an API key
`GET`	`/api/github/repos`	List user's GitHub repos
`POST`	`/api/repos/connect`	Schedule background clone+index

---

20. MCP Server — All 21 Tools

The MCP (Model Context Protocol) server runs over **stdio/JSON-RPC** and allows external AI tools (Claude Code, Cursor, Windsurf, GitHub Copilot) to query the engine programmatically.

#	Tool	Inputs	Returns
1	`search_code`	`query`, `limit`, `lang`	Hybrid BM25+HNSW search results
2	`find_callers`	`name`	All functions that call this symbol
3	`find_callees`	`name`	All functions called by this symbol
4	`get_endpoints`	—	All detected HTTP API routes
5	`get_file_structure`	`file`	AST structure of a specific file
6	`find_similar`	`code`, `limit`	Semantically similar code snippets
7	`get_stats`	—	Index summary (chunk count, file count)
8	`get_god_nodes`	`limit`	Highest-degree nodes in the call graph
9	`get_communities`	—	Label-propagation clusters
10	`get_surprises`	—	Cross-community unexpected edges
11	`explain`	`question`, `context_chunks`	LLM RAG explanation with code context
12	`get_health_score`	—	Codebase health score (0-100)
13	`get_security_scan`	—	Secrets, XSS, SQLi, hardcoded passwords
14	`blast_radius`	`file`	Dependency depth/risk for a file change
15	`analyze_legacy_code`	`file`	Refactoring plan for legacy code

#	Tool	Inputs	Returns
16	`find_dead_code`	—	Unreachable functions (0 incoming edges)
17	`isolate_microservice`	`cluster_id`	Cross-boundary interfaces for carving a community
18	`get_obligations`	`doc`	Ingested policy obligations
19	`check_compliance`	`obligation_id`	Compliance status for a specific obligation
20	`get_policy_gaps`	`doc`	Unmapped policy obligations
21	`get_compliance_report`	`format`	Full audit report across all policies

---

21. Tauri Desktop Application

Configuration (`src-tauri/tauri.conf.json`)

Setting	Value
Product Name	Sentinel Auditor
Version	0.1.0
Identifier	`dev.somil.sentinel`
Window Title	Sentinel Auditor
Window Size	1400 × 900, centered
Dev Server	`http://localhost:5173` (Vite)
Production Frontend	`../ui/dist`
Bundled Resources	`../target/release/sentinel.exe` → `sentinel.exe`
Icon Sizes	32x32, 128x128, .ico, .icns

Setting	Value
CSP	Disabled (`dangerousDisableAssetCspModification: true`)
`withGlobalTauri`	`true` (exposes `window.__TAURI__`)

How It Works

1. Tauri spawns a native OS window (WebView2 on Windows, WebKit on macOS/Linux).
2. The frontend (React + Vite) communicates with the Rust backend via Tauri's IPC bridge.
3. The Rust backend starts the `sentinel serve` HTTP server internally.
4. The frontend fetches data from `localhost:7700/api/*`.
5. For file/folder selection, it uses Tauri's `dialog.open()` API to invoke the native OS file picker.

---

22. Frontend — All Views

The frontend (`ui/src/App.tsx`) provides four primary tabs:

Tab	View	Features
Index	Folder picker + indexing	Select local folder via native OS dialog, start indexing, see AST progress animation, success state
Search	Hybrid search interface	Text input, rich code snippet results with highlighted line numbers, file paths, similarity scores, "Find Similar" buttons
Graph	D3.js force-directed topology	Interactive call graph, zoom/pan, community coloring, sub-toggles for God Nodes / Communities / Surprises / Security Issues
Auditor	AI chat interface	Chat with local Ollama LLM about the codebase, air-gapped Sentinel AI mode

Additional UI Features (in the embedded `ui/index.html`)

- **Health Score Dashboard**: 0-100 codebase health breakdown
- **Security Scan Panel**: Lists hardcoded secrets, XSS patterns, SQL injection risks

- **Blast Radius Viewer**: Visualize the impact of changing a specific file
- **Git Churn Analytics**: Most-frequently-changed files
- **Audit Report View**: Policy compliance results with severity badges
- **Settings Panel**: Ollama URL configuration, search parameters, theme toggle

---

23. VS Code Extension

Located in `needle-vscode/`.

Commands

Command	Description
`needle.showGraph`	Open the Knowledge Graph in a Webview
`needle.openInBrowser`	Open the full UI in the default browser

Configuration

Setting	Default	Purpose
`needle.serverUrl`	`http://localhost:7700`	URL of the running Sentinel server

Activity Bar

A custom icon in the VS Code Activity Bar opens a **Search** webview panel that talks to the Sentinel API.

---

24. CLI — Complete Command Reference

Command	Subcommands	Description
`sentinel init <path>`	—	Index a directory

Command	Subcommands	Description
`sentinel search <query>`	—	Hybrid search from the terminal
`sentinel status`	—	Show index statistics
`sentinel reindex`	—	Force full re-index
`sentinel config`	—	View/edit configuration
`sentinel bench`	—	Run performance benchmarks
`sentinel watch`	—	Start live file watcher
`sentinel mcp`	—	Start MCP server (stdio)
`sentinel serve`	—	Start HTTP server + web UI
`sentinel report`	—	Generate codebase report
`sentinel graph`	—	Export code graph
`sentinel policy ingest <path>`	`--name`, `--version`, `--dry-run`, `--heuristic-only`, `--format`	Ingest a policy document
`sentinel policy list`	`--format`, `--verbose`	List ingested policies
`sentinel doctor`	`--sovereign`, `--offline-strict`, `--ollama-url`, `--ledger-path`, `--json`	Diagnostics and readiness check
`sentinel ledger append`	`--report`, `--type`, `--key`, `--gen-key-if-missing`	Append an audit block
`sentinel ledger verify`	`--ledger`, `--verbose`	Verify ledger integrity

Command	Subcommands	Description
`sentinel ledger keygen`	`--output-dir`, `--force`	Generate Ed25519 keypair
`sentinel audit`	`--ledger`, `--output`, `--json`	Run policy compliance audit

Doctor Checks

The `doctor` command verifies: 1. Compile features (zero cloud crates in sovereign mode) 2. Cloud routes disabled 3. Environment hygiene (no cloud API keys set) 4. Local Ollama reachable via raw TCP probe 5. Ledger chain integrity 6. Local storage accessibility

---

25. Memory Optimization & Latency

Why It's Fast

Optimization	Impact
No network overhead	Zero HTTP round-trips to external APIs
No external DB	No IPC or socket overhead to Qdrant/PostgreSQL
No ONNX/ML model	Hash-projection embedding takes 0.00ms vs 50-100ms for transformers
Flat embedding storage	`Vec<f32>` is contiguous in memory → CPU cache friendly
Bincode serialization	~10x faster than JSON for index load/save
Rayon parallelism	File chunking runs across all CPU cores via `par_iter`
xxH3 deduplication	Unchanged files are skipped entirely on re-index
HNSW diversity heuristic	Better search quality with fewer nodes visited

Measured Benchmarks

Metric	Value
Average total query time	**0.3 ms**
Average BM25 time	0.04 ms
Average HNSW time	0.25 ms
Average embedding time	0.00 ms
Average RRF fusion time	0.01 ms
Binary size (release)	~21 MB
Cold-start time	~125 ms
Precision	90%
Recall	90%
F1 Score	90%

(Benchmarked with model `qwen2.5-coder:7b-q4_0` on Windows)

---

26. Build System, Features & Profiles

Feature Flags

```
[features]
default = ["cloud"]
cloud = ["axum", "tower-http", "open", "sqlx", "tower-cookies", "time", "urlencoding
sovereign = []
```

Build Command	Result
`cargo build --release`	Full cloud+sovereign binary with all features
`cargo build --release --no-default-features --features sovereign`	Air-gapped binary, zero networking code

Release Profile

```
[profile.release]
opt-level = 3        # Maximum optimization
lto = true           # Link-Time Optimization (whole-program)
codegen-units = 1    # Single codegen unit for maximum optimization
```

Binary Targets

Name	Path	Purpose
`sentinel`	`src/main.rs`	Main CLI binary
`hnsw_bench`	`benches/hnsw_bench.rs`	HNSW performance benchmarks
`bm25_bench`	`benches/bm25_bench.rs`	BM25 performance benchmarks
`embedding_bench`	`benches/embedding_bench.rs`	Embedding performance benchmarks

---

27. Error Handling Strategy

All errors flow through a single `Error` enum (in `src/error.rs`) using `thiserror`:

Variant	When It Fires
`Io(std::io::Error)`	File read/write failures
`InvalidPath(String)`	Non-existent or inaccessible paths

Variant	When It Fires
`IndexNotFound(String)`	No `.needle/` index exists
`ChunkingError(String)`	Tree-sitter parse failure (after fallback also fails)
`EmbeddingError(String)`	Embedding generation failure
`IndexError(String)`	BM25/HNSW corruption
`QueryError(String)`	Search pipeline failure
`ConfigError(String)`	Invalid configuration
`SerializationError(String)`	JSON/Bincode parse failure
`PolicyError(String)`	Policy ingestion issues (scanned PDF, empty doc)
`LedgerError(String)`	Tamper detection, missing keypair, append failure
`OfflineStrictViolation(String)`	Non-loopback URL detected in strict mode
`DoctorError(String)`	Diagnostics check failure
`Other(Box<dyn Error>)`	Catch-all for unexpected errors

Rule: No `unwrap()`, `expect()`, or `panic!()` on user-input paths. All user-facing operations return `Result<T, Error>`.

---

28. Testing Infrastructure

Architecture

A comprehensive **4-tier opaque-box** test suite in `tests/e2e_sentinel_tests.rs`.

Tier	Focus	Example Tests
Tier 1	Feature coverage	Sovereign build validation (`cargo tree` checks), policy parsing (rejecting scanned PDFs), graph linking, JSON canonicalization, tamper localization

Tier	Focus	Example Tests
Tier 2	Boundary cases	Timeouts, empty documents, offline-strict URL obfuscation rejection, corrupt PDF binary resilience, concurrent file locks on ledger
Tier 3	Integration scenarios	End-to-end audit flow (ingest → index → audit → ledger sign → verify)
Tier 4	Adversarial hardening	Tampered payload single-char alteration, deleted middle block, sequence gap injection, signature forgery

Test Fixtures

Fixture	Location	Purpose
`tests/fixtures/policies/security_standard_v1.md`	Policy document	Standard security policy for compliance testing
`tests/fixtures/ledgers/empty_chain.jsonl`	Empty ledger	Verifies clean verification of empty chain
`tests/fixtures/ledgers/valid_three_block_chain.jsonl`	Valid 3-block chain	Verifies correct chaining
`tests/fixtures/ledgers/tampered_payload_seq1.jsonl`	Tampered block 1	Verifies tamper detection at exact sequence
`tests/fixtures/ledgers/tampered_sequence_gap.jsonl`	Missing sequence	Verifies sequence gap detection
`tests/fixtures/ledgers/tampered_prev_hash.jsonl`	Broken hash link	Verifies prev_hash mismatch detection

Fixture	Location	Purpose
`tests/fixtures/ledgers/tampered_signature.jsonl`	Forged signature	Verifies signature verification failure

---

29. Verified Benchmarks

See `docs/BENCHMARKS.md` for the full data.

> **Important**: These numbers were captured on the dev machine using Windows PowerShell scripts with model `qwen2.5-coder:7b-q4_0`. They must be re-run on the actual demo machine close to Aug 25.

---

30. Complete Feature List

#	Feature	Status
1	Sovereign Build Mode (`--features sovereign`)	✅ Built
2	`sentinel doctor --sovereign` diagnostics	✅ Built
3	Zero-network dependency guarantee	✅ Verified via `cargo tree`
4	Default build backward compatibility	✅ Verified
5	Local-only LLM routing (Ollama loopback)	✅ Built
6	`--offline-strict` enforcement	✅ Built
7	PDF/Markdown/text policy parser	✅ Built
8	Scanned-image PDF guard	✅ Built
9	LLM + rule-based obligation structuring	✅ Built
10	Policy-code compliance matching	✅ Built

#	Feature	Status
11	Policy compliance graph	✅ Built
12	`sentinel audit` CLI command	✅ Built
13	MCP compliance tools	✅ Built
14	Canonical JSON block encoding	✅ Built
15	SHA-256 hash-chained blocks	✅ Built
16	Ed25519 digital signatures	✅ Built
17	Redacted keypair security	✅ Built
18	Append-only JSONL ledger writer	✅ Built
19	Fresh/empty chain clean verification	✅ Built
20	Tamper detection + exact localization	✅ Built
21	Tree-sitter AST chunking (7 languages)	✅ Built
22	Hand-rolled BM25 inverted index	✅ Built
23	Hand-rolled HNSW vector index	✅ Built
24	Hash-projection 384-dim embeddings	✅ Built
25	Reciprocal Rank Fusion	✅ Built
26	CodeGraph with communities + god nodes	✅ Built
27	D3.js force-directed graph visualization	✅ Built
28	Tauri v2 native desktop app	✅ Built
29	21-tool MCP server	✅ Built
30	VS Code extension	✅ Built
31	File watcher (live re-indexing)	✅ Built
32	Legacy language fallback (COBOL demo)	✅ Built

31. Onboarding FAQ for New Developers

Q: Why is the crate named `needle` but the binary is `sentinel`? A: The internal library crate retains the name `needle` to avoid massive refactoring risk across all `use needle::*` imports. Only user-facing surfaces (binary, window title, README, docs) are branded as "Sentinel Auditor".

Q: Why did we hand-roll HNSW and BM25 instead of using a crate? A: To minimize binary bloat and eliminate external service dependencies. By hand-rolling, the entire engine is a single ~21MB executable. No Docker containers, no background database services, no network connections. It starts in 125ms.

Q: How do embeddings work without a neural model? A: We use hash-projection (a form of random projection rooted in the Johnson-Lindenstrauss lemma). xxH64 hashes act as pseudo-random projections, scattering token features across a 384-dim vector. L2 normalization ensures dot product equals cosine similarity. This gives us semantic grouping at 0.00ms per chunk.

Q: What if the codebase uses a language we don't support (COBOL, Fortran)? A: Tree-sitter gracefully falls back. If it can't parse the syntax, the chunker switches to a 60-line sliding window with 15-line overlap. It never crashes or skips files. The COBOL file in `demo/legacy_system.cobol` is the live proof.

Q: Where does the LLM run and what model do we use? A: Locally via Ollama (`localhost:11434`). Default model is `qwen2.5-coder:7b-q4_0`. The `--offline-strict` flag explicitly rejects any non-loopback connection. In sovereign mode, even the HTTP client library (`reqwest`) is compiled out — we use raw TCP sockets to talk to localhost only.

Q: How do I run the project from scratch? A: `cargo build --release`, then `./target/release/sentinel init <your-code-directory>` to index. `sentinel serve` starts the web UI at `http://localhost:7700`. `sentinel mcp` starts the MCP server for IDE integration.

Q: How do I add a new language to Tree-sitter support? A: Add the `tree-sitter-<language>` crate to `Cargo.toml`, add a `Language::NewLang` variant to `schema.rs`, add AST node types to the match block in `chunking/code.rs`, and add the file extension mapping.

Q: How is the ledger tamper-proof? A: Each block contains the SHA-256 hash of the previous block (`prev_hash`), forming a chain. The payload is canonically serialized to eliminate key-ordering nondeterminism, then hashed. The entire block is signed with Ed25519. Modifying any

block breaks the chain at the exact point of tampering, and the verifier reports the specific sequence number.

Q: What data leaves my machine? A: In sovereign mode: **nothing**. The build flag removes all networking code at compile time. In cloud mode: only if you explicitly connect a GitHub repo (which triggers a `git clone` via your OAuth token). Code is never sent to any third-party API in either mode (LLM calls go to localhost only in sovereign mode).

This document was compiled from a deep read of every source file in the Sentinel Auditor codebase at `d:\AEGIS_AST` as of August 15, 2026.